

CSE 4351: Computer Security



Instructor: Dr. MD Armanuzzaman

Biometric Authentication

- **Biometric authentication systems** authenticate an individual based her physical characteristic
- Types of biometric used in authentication
 - face
 - palm geometry
 - fingerprint
 - Iris
 - Signature
 - Voice
- Most common uses of biometric authentication is for specific applications rather than computer authentication

Biometric Authentication

- Like other authentication mechanisms, biometric authentication includes an enrollment phase during which a biometric is captured
 - the initial reading is often called a template
 - at authentication time, a new biometric reading is performed and is compared to the stored template
- Unlike other authentication mechanisms, biometric **matching is approximate**
 - each reading can be influenced by a variety of factors
 - e.g., light conditions, facial expressions, hair style, glasses, etc. for face recognition
 - some types of biometrics can match more accurately than others
 - e.g., iris vs. face or palm

Biometric Authentication

- Biometric matching can be used to perform
 - verification
 - user's biometric scan is used to match her own template only
 - identification
 - user's biometric scan is used to match a database of templates
- Identification might not always be possible
- Biometric systems attempt to minimize
 - false reject rate: authentic biometric is rejected
 - false accept rate: imposter biometric is accepted
- Depending on the environment, minimizing one of them might be more important than minimizing both

Biometric Authentication

- **New types of biometrics** are being explored
 - brain waves, heart beats, etc.
- Many forms of traditional biometrics can be stolen
- Static biometrics can be replayed

Biometric Authentication

- Current research direction: **biometric key generation**
 - the idea: a biometric can be used to generate a cryptographic key
 - the key can be reproduced using another biometric close enough to the original
 - no need to remember any information such as a password
 - the key can be used for authentication or encryption
 - key generation algorithm produces a helper data that can later aid in recovering the same key from a noisy version of the biometric
 - security requirements are strict
 - the helper data must leak minimal information about the biometric
 - compromise of the key must not lead to recovery of the biometric

Summary

- **Entity authentication** is an important topic with the main application in access control
- **Various techniques exist** ranging from time-invariant passwords to provably secure identification schemes
- Despite the weak security password-base authentication provides, it is the **most widely used authentication mechanism**
 - ease of use, user familiarity, no infrastructure requirements

Liveness is Not Enough: Enhancing Fingerprint Authentication with Behavioral Biometrics to Defeat Puppet Attacks

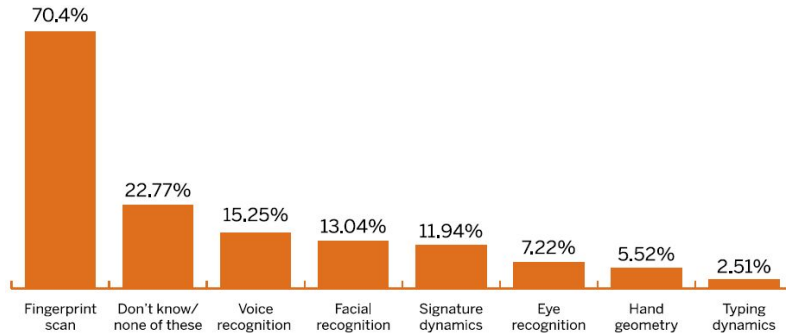


USENIX Security' 20

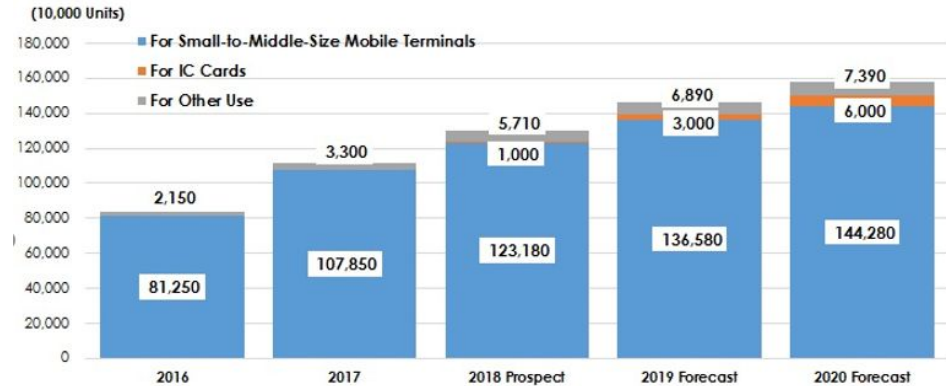
The Prevailing Fingerprint Authentication



PHOTO: <https://www.fingerprints.com/>



German, R. L., & Barber, K. S. (2018). Consumer attitudes about biometric authentication. *The University of Texas at Austin*.



Report: Yano Research Institute Ltd.

Attacks on Fingerprint Authentication

ICS > 35 > 35.240 > 35.240.15

ISO/IEC 30107-1:2016

Information technology — Biometric presentation attack detection — Part 1: Framework



Author:
Lindsey O'Donnell

April 8, 2020 / 9:00 am

15:30 minute read

New research used 3D printing technology to bypass fingerprint scanners, and tested it against Apple, Samsung and Microsoft mobile products.

New research has found that it's possible to use 3D printing technology to create "fake fingerprints" that can bypass most fingerprint scanners used by popular devices. But, creating the attack remains costly and time-consuming.

Researchers with Cisco Talos created different threat models that use 3D printing technology, and then tested them on **mobile devices** (including the iPhone 8 and Samsung S10), laptops (including the Samsung Note 9, Lenovo Yoga and HP Pavilion X360) and smart devices (such as a smart padlock).

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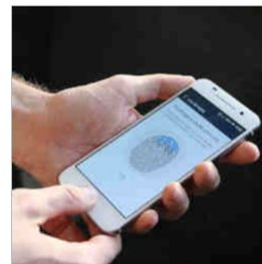


ACM NEWS

Attackers Can Bypass Fingerprint Authentication with an ~80% Success Rate

By Ars Technica
April 8, 2020
[Comments](#)

VIEW AS: SHARE:



For decades, the use of fingerprints to authenticate users to computers, networks, and restricted areas was mostly limited to large and well-resourced organizations that used specialized and expensive equipment. That all changed in 2013 when Apple introduced TouchID. Within a few years, fingerprint-based validation became available to the masses as computer, phone, and lock manufacturers added sensors that gave users an alternative to passwords when unlocking the devices.

Although hackers managed to [defeat TouchID with a fake fingerprint](#) less than 48 hours after the technology was rolled out in the iPhone 5, fingerprint-based authentication over the past few years has become much harder to defeat. Today, fingerprints

Puppet Attack

Police 'visit funeral home to unlock dead man's phone'

© 23 April 2018



Police in Florida have been criticised for allegedly entering a funeral home in a futile bid to unlock a dead man's smartphone.

6-year-old uses sleeping mom's thumb to go on Amazon shopping spree

by WKRC | Wednesday, December 28th 2016



6-year-old uses sleeping mom's thumb to go on Amazon shopping spree (Provided by/edited with permission: Bethany Johnson Howell)

Child uses sleeping mom's fingerprints to buy Pokemon gifts

When you want to buy \$250 worth of Pokemon presents, desperate times call for desperate measures.



Alfred Ng Dec. 27, 2016 6:25 a.m. PT



I got drunk last night and got robbed because I was using Touch ID :-)

laDouche

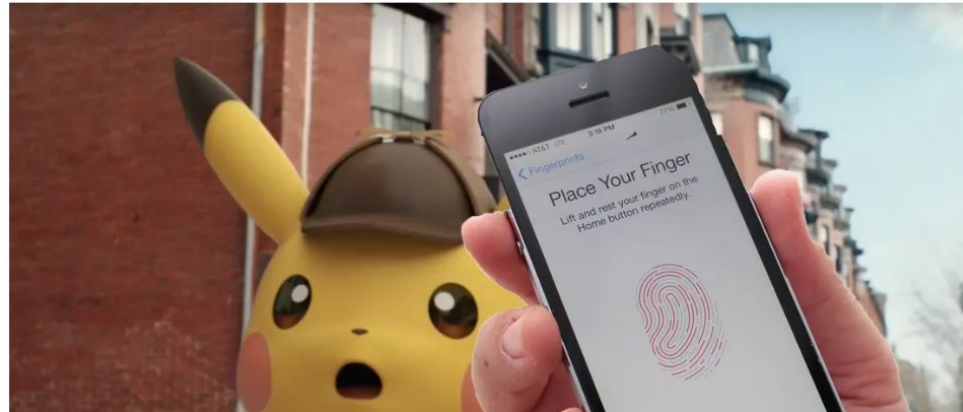
December 2014 edited December 2014



hi guys,

not looking to blame anyone but thought i'd share my tale of sorrow here...

long story short, i was at a party last night and i passed out after some heavy drinking. i woke up this morning and walked to an atm machine wanting to get some cash out for a cab. to my amazement, the transaction was declined. so i whipped out my shiny new iphone 6, fired up 1password, placed my thumb for the touchid, and logged in to my online banking website.



Puppet Attack

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© 23 April 2018

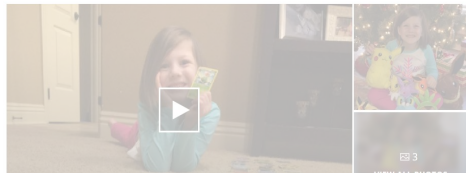
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Existing liveness detection methods all fail in defeating puppet attacks.



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laDouche

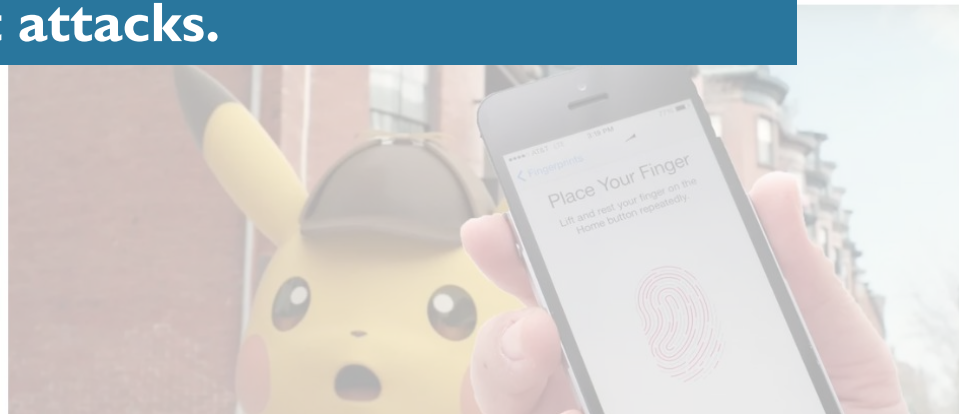
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Approach



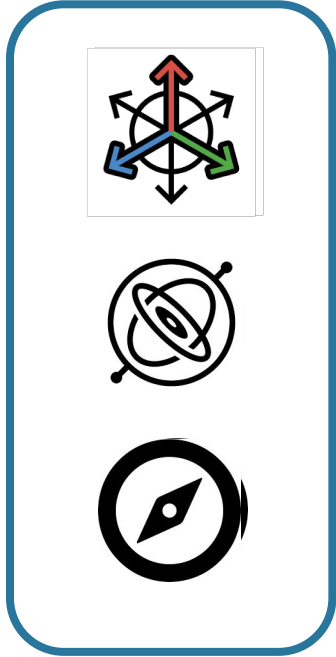
Fingerprint



Fingertip-touch
behavior

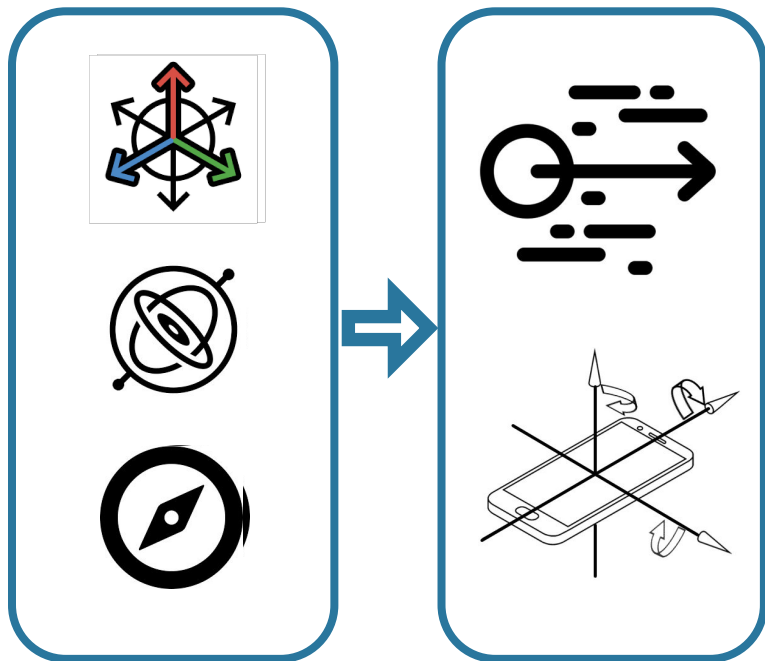
**Complement fingerprint authentication with
fingertip-touch behavioral characteristics**

System Overview



Data capture

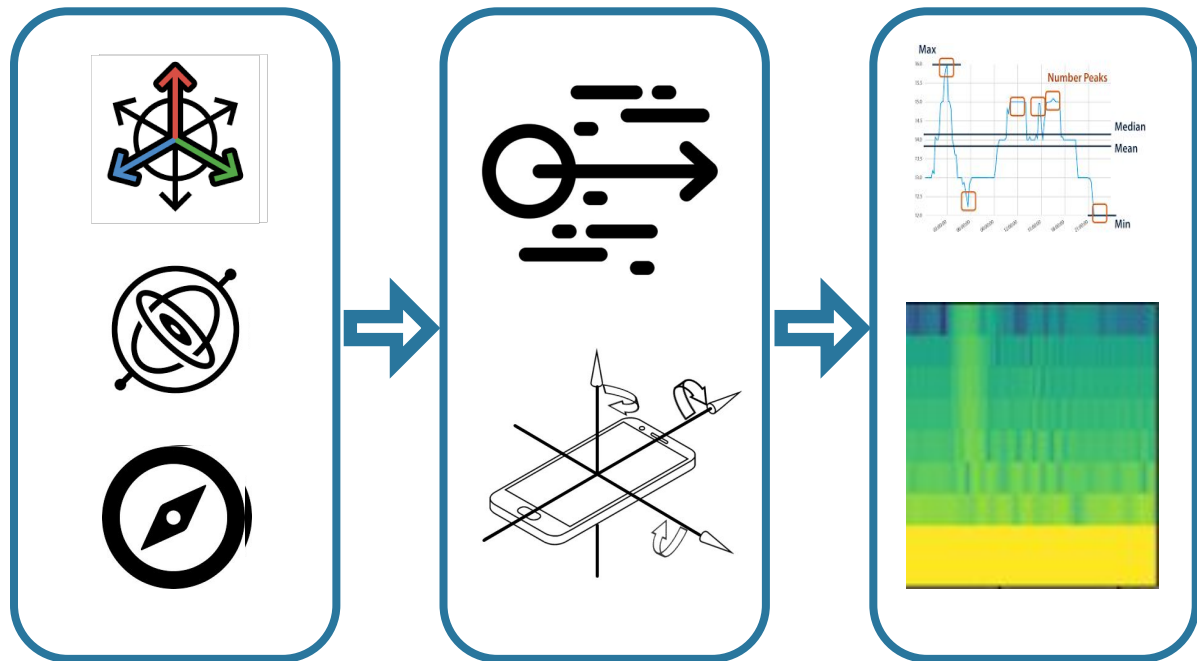
System Overview



Data capture

Behavior
characterizing

System Overview

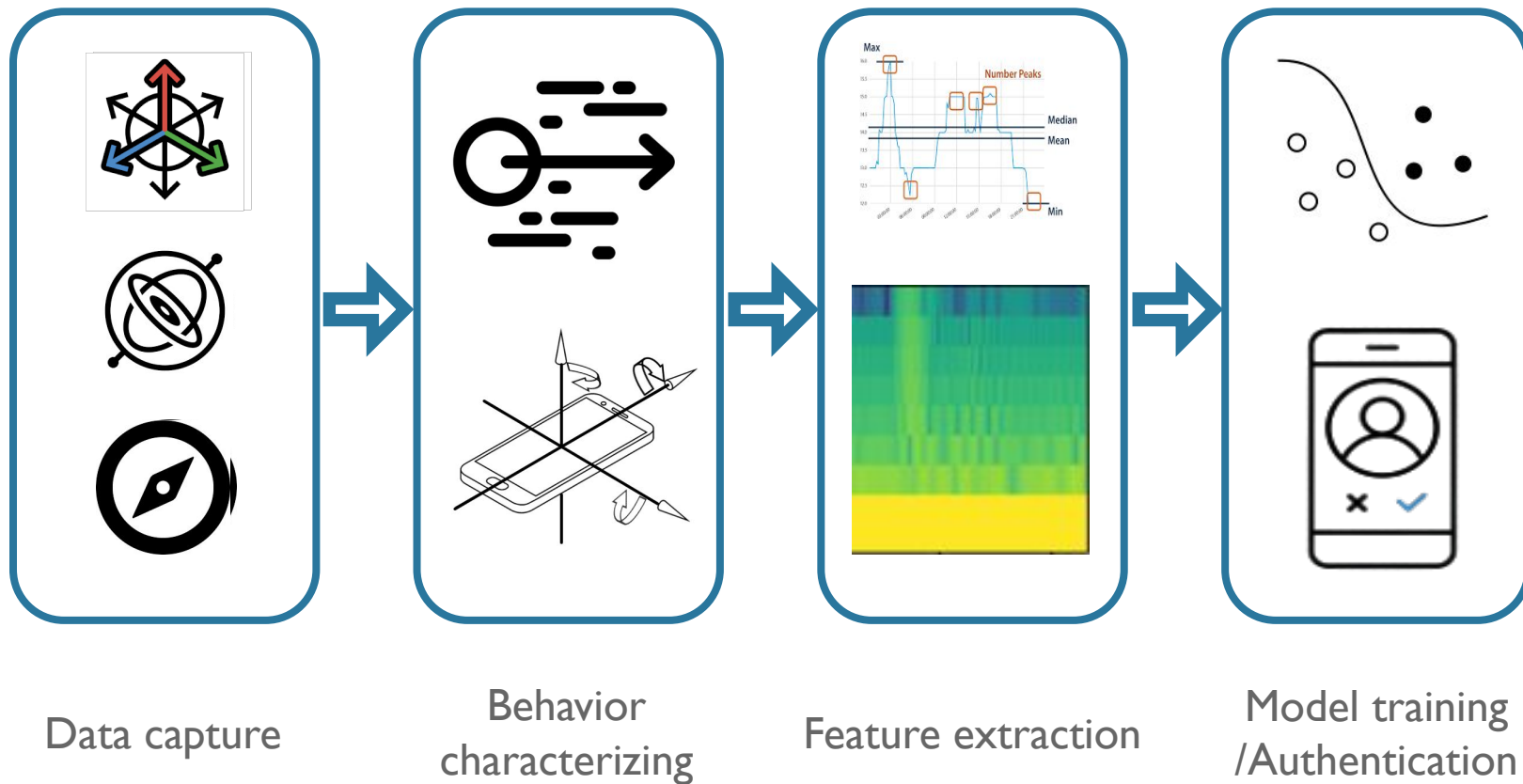


Data capture

Behavior
characterizing

Feature extraction

System Overview



Time- and Frequency- Domain Features (TFF)

Table 1: Time- and frequency-domain features and their normalized fisher’s scores.

Domain	Feature	Description	Normalized Fisher Score of ($a_x, a_y, a_z, a', \phi, \theta, \psi$)
Time	Mean	The mean of the time series.	(0.45, 0.01, 0.22, 0.68, 0.86, 0.84, 0.84)
	Standard deviation	The standard deviation of the time series.	(0.24, 0.56, 0.31, 0.41, 0.58, 0.32, 0.74)
	Relative standard deviation	The extent of variability in relation to its mean.	(0.34, 0.15, 0.12, 0.56, 0.71, 0.64, 0.82)
	Sum of absolute differences	The sum over the absolute value of consecutive changes in the time series.	(0.32, 0.27, 0.72 , 0.52, 0.53, 0.72, 0.78)
	Absolute energy	The absolute energy of the time series.	(0.63, 0.98, 0.85 , 0.57, 0.72 , 0.57, 0.37)
Frequency	Autocorrelation	The autocorrelation of the time series.	(0.00, 0.14, 0.15, 0.21, 0.94, 0.62, 0.64)
	Spectral centroid	The center of mass of the spectrum is located.	(0.34, 0.21, 0.38, 0.12, 0.78, 0.98, 0.78)
	Spectral spread	The average spread of the spectrum in relation to its centroid.	(0.66 , 0.36, 0.32, 0.78 , 0.46, 0.82, 0.96)
	Spectral skewness	The measurement of the asymmetry of the probability distribution of a real-valued random variable about its mean.	(0.85 , 0.45, 0.58, 0.84 , 0.56, 0.85, 1.00)
	Spectral kurtosis	The shape of a probability distribution.	(0.34, 0.17, 0.70, 0.86, 0.62 , 0.51, 0.42)
	Power spectral density	Average of distribution of power into frequency components.	(0.90, 0.71, 0.86 , 0.26, 0.85, 0.68, 0.82)
	Spectral entropy	The complexity of the signal in the frequency domain.	(0.94 , 0.32, 0.82 , 0.21, 0.96, 0.82, 0.89)

CNN-based Features (CNF)

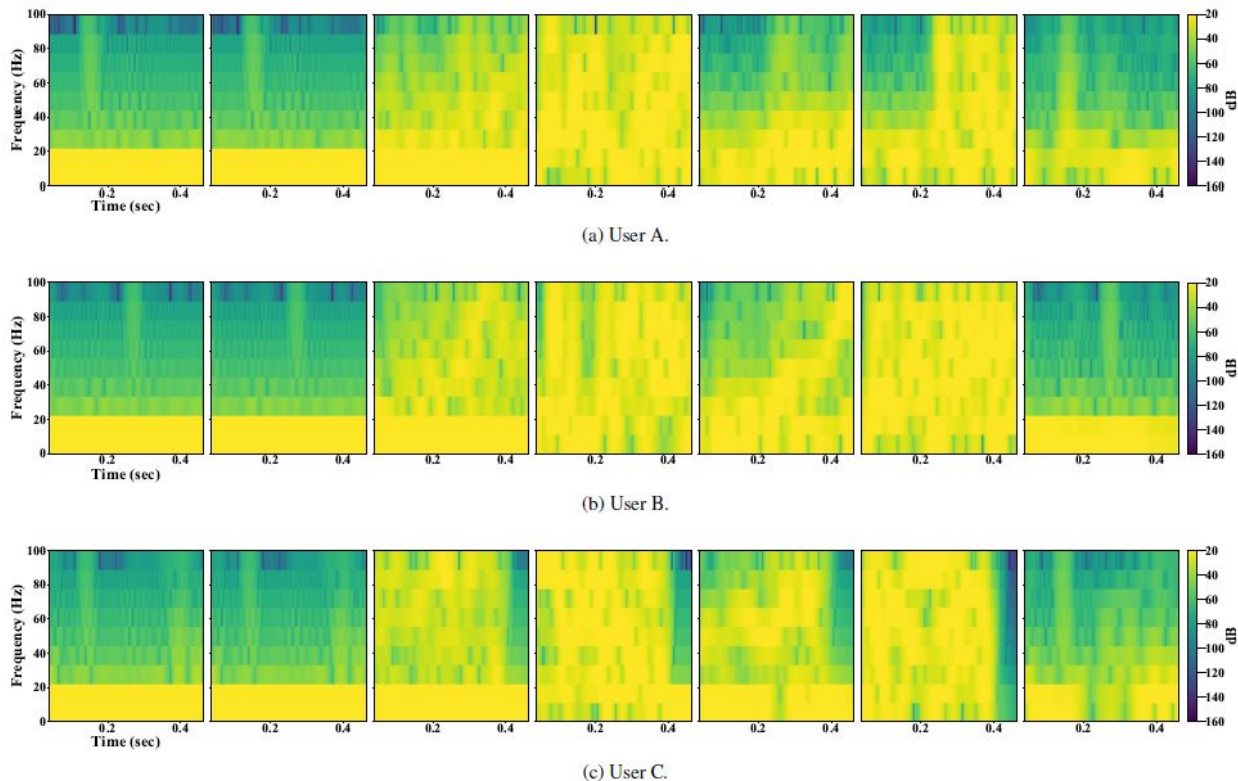
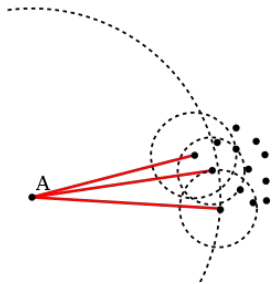


Figure 3: Characterized fingertip-touch behaviors of three users under STFT. From left to right, spectrograms of a_x , a_y , a_z , a' , θ , ϕ , ψ .

One-class Classifier

$$r_{XY} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

Pearson coefficient-based similarity
comparison (PCC)

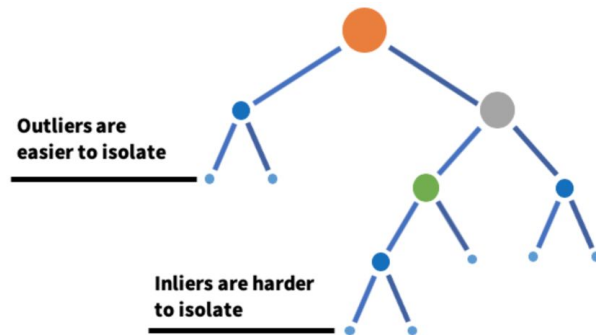


Local outlier factor (LOF)

$$\min_{R,a} R^2 + C \sum_{i=1}^n \zeta_i$$

$$s. t. \|x_i - a\|^2 \leq R^2 + \zeta_i, i = 1, \dots, n$$
$$\zeta_i \geq 0, i = 1, \dots, n$$

在此处键入公式。One-class support vector
machine (OCSVM)



Isolation forest (IF)

Data Collection

Table 3: Summary of the compiled datasets

Dataset	Week of Collection	# of Subjects / Attackers	Postures	Device	# of Data Points
1	1 †, 8 and 9 ‡	90	Sitting, standing, lying, walking, running	OnePlus3	63,000
2A	2, 3, 5, 7	24, 24, 22, 21	Sitting	OnePlus3	18,200
2B	10, 11, 12, 13	62, 61, 59, 53			47,000
3	Added Aug. 2019	64	Sitting	Xperia XZ1, Oneplus5, Vivo X21	3,200
4A	2 †, 10 and 11 ‡	15	Sitting	OnePlus3	3,600
4B					3,600
4C					3,600

Datasets

- 90 subjects in the data collection.
- Compiled three datasets in different postures¹, periods², and devices³.
- Compiled one attack dataset⁴ by considering three attacks with 15 subjects as adversaries.

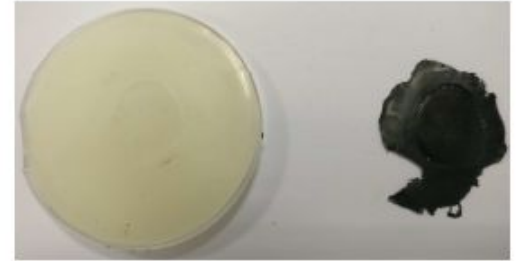


Figure 4: Artificial fingerprint replica. The left is the mold used to capture fingerprint; the right is a fake fingerprint crafted using silicone rubber.

Reliability Evaluation

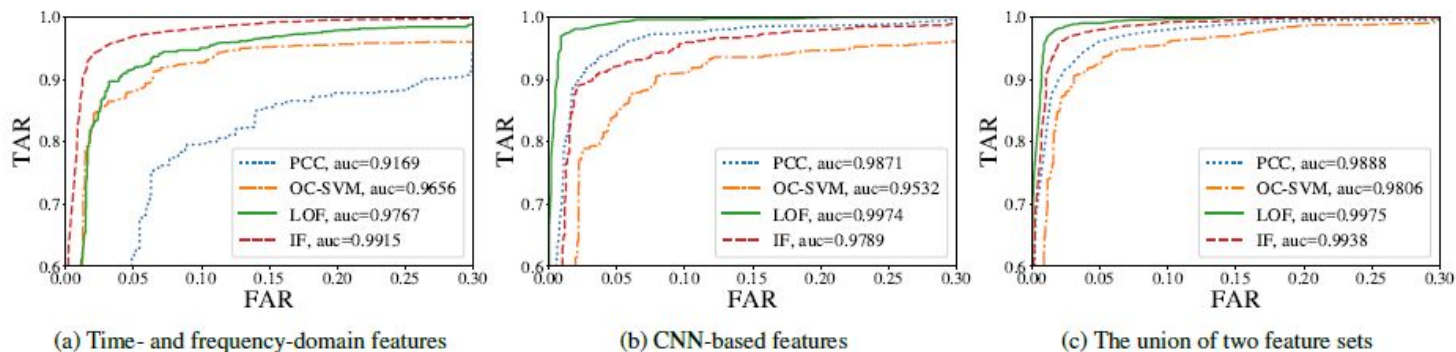


Figure 5: ROC curves of different feature sets under different one-class classifiers.

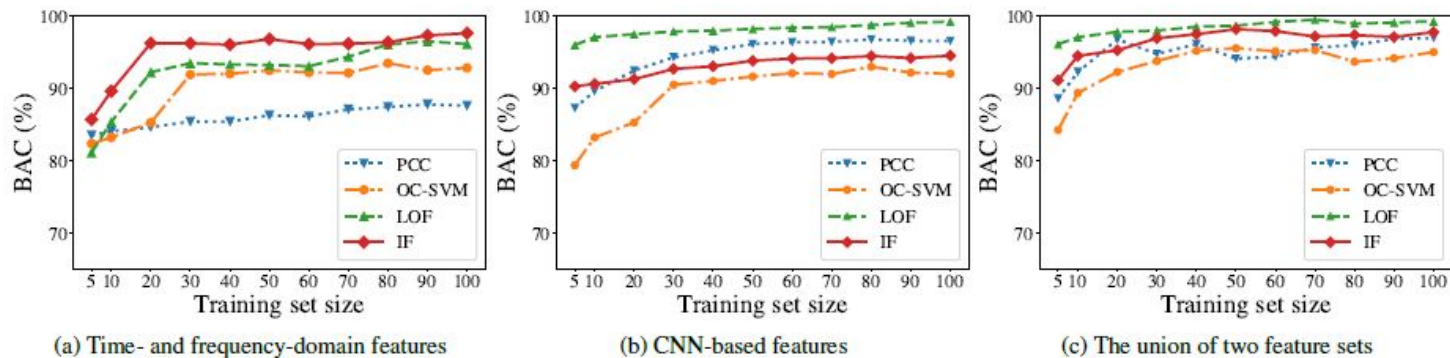


Figure 6: BAC under different classifiers and different feature sets at varying training set sizes.

Reliability Evaluation

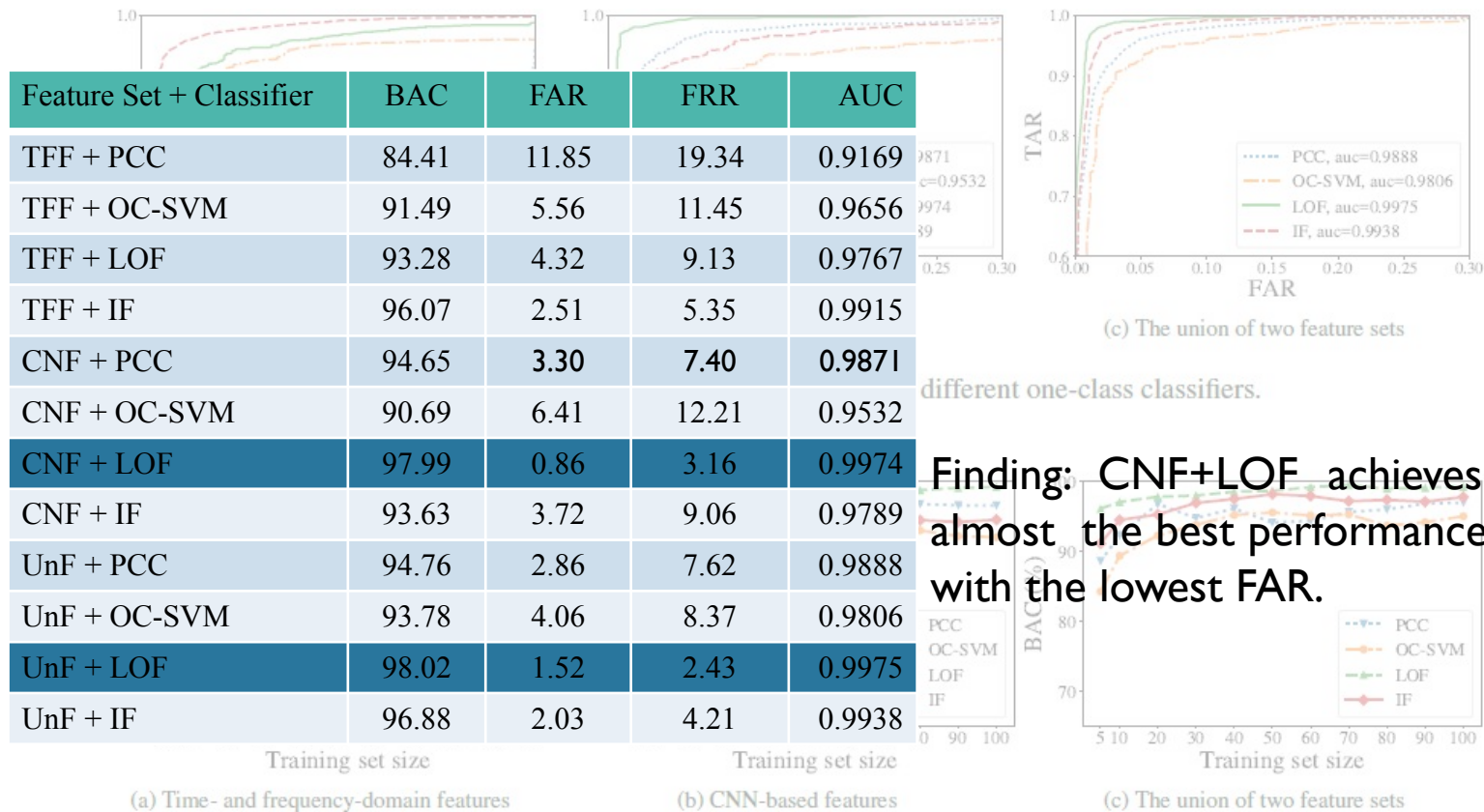
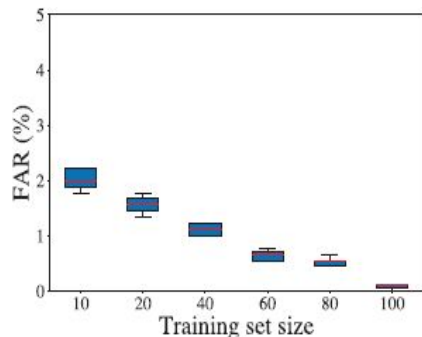
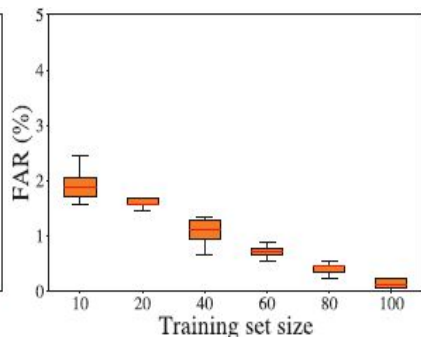


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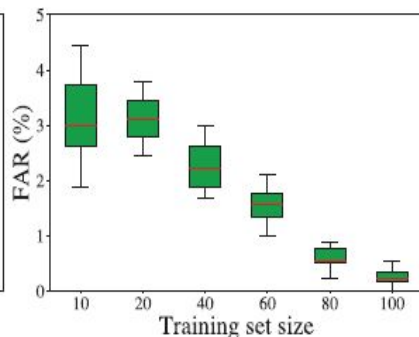
Evaluation of Presentation Attacks



(a) Artificial replica attack



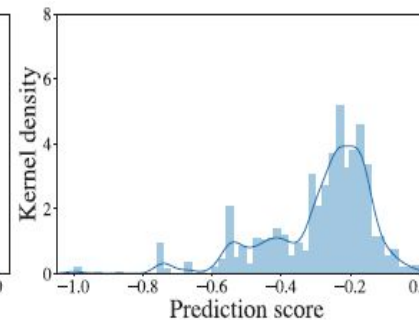
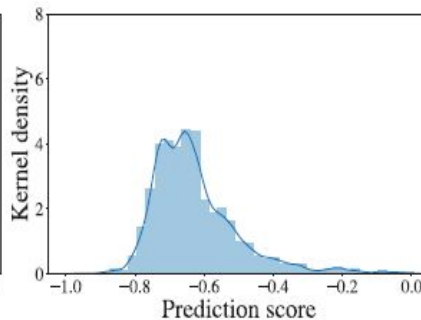
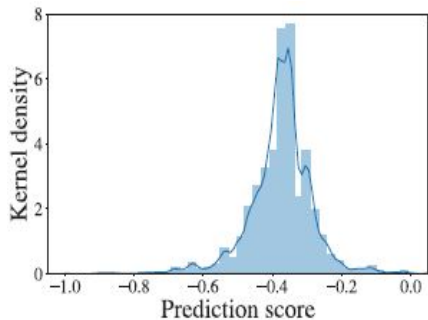
(b) Puppet attack



(c) Mimicry attack

Mean/standard deviation of FAR and prediction score

Attack	FAR	Score
ARA	0.08/0.06	-0.29/0.15
PA	0.12/0.08	-0.62/0.13
MA	0.25/0.14	-0.37/0.10



FAR and kernel density of prediction score under attacks

Limitations

Behavior variability with
time elapsing?

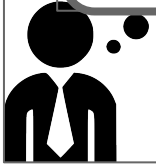


PHOTO: iconfinder.com



PHOTO:
<https://parker-marker.com/>

